

SYLLABUS :-

Linear algebra, or, more specifically, the study of linear vector spaces, have come up over years as the foundation for many subjects in Electronics and Communication Engg. like signal processing, communications, pattern recognition, machine learning, image processing, control, even VLSI design. Many algorithms in the above subjects are derived, analyzed and also visualized using advanced vector space concepts like spectral theory, subspace decomposition, matrix factorization, orthogonal projection etc. In many reputed universities overseas and also in some frontline IITs, the ECE and likewise departments offer courses on this to their own UG/PG students. Separately, last decade has seen an explosive growth in the usage of optimization techniques, in particular convex optimization concepts, in the solution of complex problems arising in electrical science subjects. As computational power became abundantly available, more and more sophisticated convex optimization techniques came into play in many real life applications. Again, the subject of optimization is built upon at the base on concepts from vector space theory. The coverage on linear algebra in the first part of this course thus helps to provide the right platform for taking a plunge into optimization theory and algorithms. Both linear algebra and optimization theory are taught in various other departments including maths in various forms. However, the orientation and focus of the subject changes from department to department. When it comes to electrical sciences, efforts should be given to orient the course suitably by using perspectives of the above stated subjects.

Recommended Text Books : 1. Convex Optimization in Signal Processing and Communications, Daniel Palomar and Yonina Eldar, Cambridge University Press, 2009. 2. Optimization Models by Giuseppe Calafiore and Laurent El Ghaoui, Cambridge University Press, 2014. 3. Linear Algebra by Hoffman and Kunze, Prentice Hall, 2015. 10. Reference Books/Materials : 1. Convex Optimization by Boyd and Vandenberghe, Cambridge University Press, 2016. 2. Linear Algebra and Its Applications by Gilbert Strang, Wellesley